SIT330-770: Natural Language Processing

Week 0 - Course Overview

Dr. Mohamed Reda Bouadjenek

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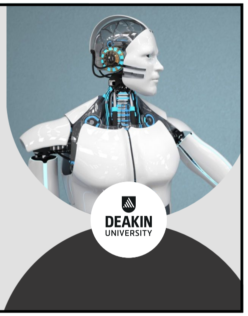

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Plan

- Introduction to SIT330-770
  - Unit team
  - Unit objectives and Introduction to NLP
  - Unit structure
  - Teaching and Assessment
  - Unit logistics

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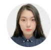
Unit team

- **Unit chair:**
  - *Reda Bouadjenek*
  - Email: reda.bouadjenek@deakin.edu.au

- **Online seminar:**
  - *Saeedeh Javadi*
  - Email: saeedeh.javadi@deakin.edu.au

- **Burwood seminars:**
  - *Ngoc Dung Huynh*
  - Email: ndhuynh@deakin.edu.au



- **Online seminar:**
  - *Echo Zhou*
  - Email: echo.zhou@deakin.edu.au

- **Online seminar:**
  - *Aymen Khouas*
  - Email: s223766761@deakin.edu.au

- **Marking tutor:**
  - *Lakpa Tamang*
  - Email: ld.tamang25@gmail.com

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Unit Chair

- Dr. Reda Bouadjenek <reda.bouadjenek@deakin.edu.au>
  - Unit chair, Waurn Ponds
- Senior Lecturer of Applied AI at Deakin University since November 2019
  - Interested in information Retrieval, Natural Language Processing, social media analysis, recommender systems
- Previously:
  - **2017-2019**: Research fellow at Toronto University, Canada
  - **2015-2017**: Research fellow at Melbourne University
  - **2014-2015**: Research fellow at inria, France
  - **2009-2013**: PhD student at the University of Paris-Saclay, France

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Unit objectives

Understanding principals behind

Natural Language Processing

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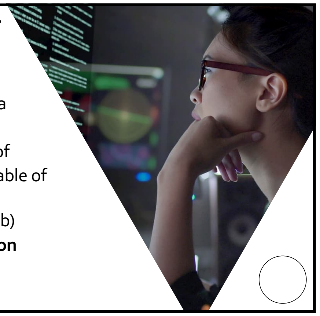
What is natural language processing?

The study of language and linguistic interactions from a computational perspective, enabling the development of algorithms and models capable of (a) **natural language understanding (NLU)** and (b) **natural language generation (NLG)**



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Go beyond the keyword matching

- Identify the structure and meaning of words, sentences, texts and conversations
- Deep understanding of broad language
- NLP is all around us

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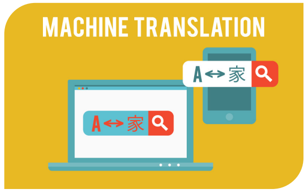
Machine translation



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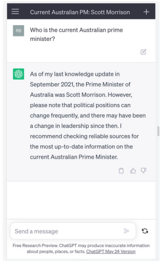
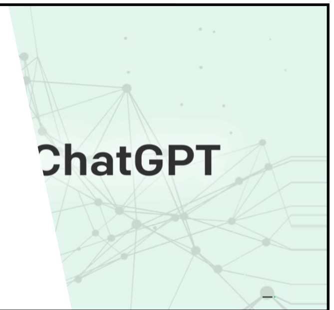
Conversational System





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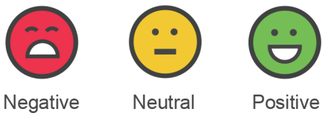
Sentiment/Opinion Analysis



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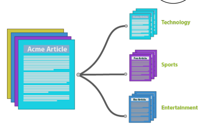
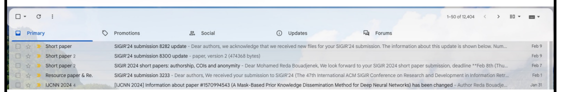
Text Classification





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Natural language instruction

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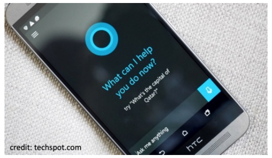
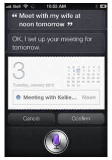
Digital personal assistant

More on natural language instruction





- Semantic parsing – understand tasks
- Entity linking – “my wife” = “Kellie” in the phone book

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Information Extraction

- Unstructured text to database entries

New York Times Co. named Russell T. Lewis, 45, president and general manager of its flagship New York Times newspaper, responsible for all business-side activities. He was executive vice president and deputy general manager. He succeeds Lance R. Primis, who in September was named president and chief operating officer of the parent.

Person	Company	Post	State
Russell T. Lewis	New York Times newspaper	president and general manager	start
Russell T. Lewis	New York Times newspaper	executive vice president	end
Lance R. Primis	New York Times Co.	president and CEO	start

Yoav Artzi: Natural language processing

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Course Outline

Module I: Foundations of NLP

- **Week 1:** Information Retrieval Part 1
- **Week 2:** Information Retrieval Part 2
- **Week 3:** Text processing

Module II: Language Modeling & Representations

- **Week 4:** N-gram Language Models
- **Week 5:** Vector Embeddings and Sequence Labeling

Module III: Advanced NLP

- **Week 6:** Neural Networks for NLP (NNs, RNNs, and Neural LMs)
- **Week 7:** Transformers and Pretrained LMs
- **Week 8:** Large Language Models (LLMs)

Module IV: Applications

- **Week 9:** Speech Processing & ASR
- **Week 10:** Dialogue Systems & Conversational AI
- **Week 11:** Wrap-up

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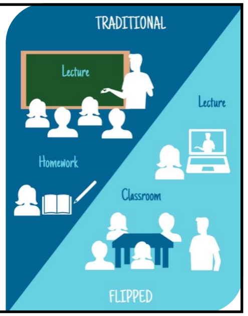
Teaching: Engaging Learning Experience

What to Expect:

- **Active Learning:** Get ready for an interactive and hands-on learning experience.
- **Flipped Classroom Model:** We're flipping the traditional classroom! Pre-class materials will be provided for self-study before our sessions.
- **Your Role:** Come prepared to discuss, question, and apply concepts during class time.
- **Collaborative Environment:** Engage with your peers, share ideas, and actively participate.



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Assessment

- OnTrack learning portfolio 100% https://ontrack.deakin.edu.au/
  - An introduction video is available on DeakinCloud
- You set your target grade (P/C/D/HD)
  - Several tasks (tailored to your target grade) need to be completed regularly
  - Tasks have different due dates
- Weekly submissions are to get feedback
  - **Maximum two feedbacks. Please review carefully your assignment before submitting it!**
- Portfolio submission at the end of the trimester is very important!
  - **Assessments will be published gradually during the trimester!**
- **Please do not put assessments on public repositories (GitHub)!**

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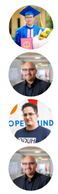
Weekly Session Details

- **SIT330**
  - Session 1:
    - Time: Monday 14:00-16:50
    - Location: Burwood B4.22
  - Session 2:
    - Time: Friday 16:00-18:50
    - Location: Burwood B4.22
  - Session 3:
    - Time: Thursday 18:00-19:50
    - Location: Online on MS Team

- **SIT770**
  - Session 1:
    - Time: Friday 11:00-13:50
    - Location: Burwood T1.13
  - Session 2:
    - Time: Tuesday 16:00-17:50
    - Location: Online on MS Teams
  - Session 3:
    - Time: Tuesday 12:00-14:50
    - Location: Waurn Pond IB4.202
  - Session 4:
    - Time: Thursday 13:00-15:50
    - Location: Waurn Pond IB4.202



Do not create a calendar meeting!
Weekly material will be gradually released!

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Communication

- Please ask questions on the discussion forum to share with others and avoid redundant inquiries
- **Please use email only if necessary**
  - You must start your email subject with **"[SIT330] or [SIT770]"**
  - I will normally answer your email within 48 hrs (working days)
  - If you don't receive a reply within 48 hrs, this often means that your question had been answered somewhere else before (e.g., in the class, forum, etc.)
  - If in doubt, check with me again during allocated contact time

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This course requires you to write substantial code in Python to achieve a C, D, or HD grade!

python™

The unit does not teach Python!

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Unit Guide

- Graduate Learning Outcomes (GLOs)
- Unit Learning Outcomes (ULOs)
- Teaching team contact details
- Learning activities
  - Class and workshop
- Reference resources
- Trimester plan
- Key dates

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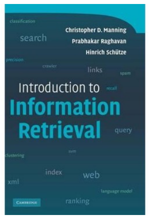
Textbook: Information Retrieval

- **Introduction to Information Retrieval**
  - *Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze*
    - Draft chapters, April 1, 2009

  - https://nlp.stanford.edu/IR-book/pdf/irbookonlinereading.pdf



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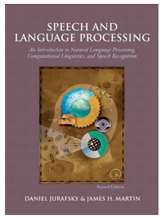
Textbook: Natural Language Processing

- **Speech and Language Processing** (3rd ed. draft)
  - *Dan Jurafsky and James H. Martin*
    - Draft chapters in progress, January 7, 2023

  - https://web.stanford.edu/~jurafsky/slp3/



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Related units

- SIT744- Deep Learning
  - Explain deep learning and its role in data science and AI

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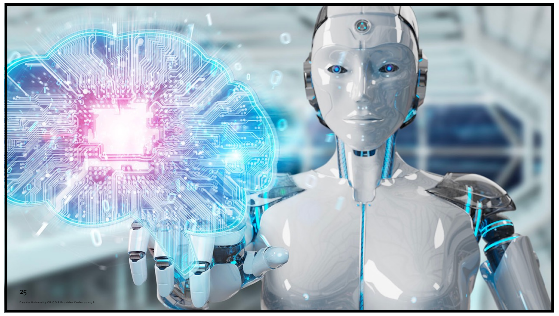

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